

Tharidu Deshan Ekanayaka

Software Engineer | Full-Stack Development and AI Systems

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SUMMARY

Full-stack engineer building production systems in Python, TypeScript, React and Node.js, with applied machine learning and LLM integration. Currently building backend AI services that put language models behind validated APIs, with every response checked against a fixed schema before a caller sees it.

WORK EXPERIENCE

FlyRank (AI search growth platform)

Jun 2026 - Present

Backend AI Engineering Intern

Remote

- Building backend AI services in **Node.js** and **Express** through a structured engineering track, from **PostgreSQL** persistence and Supabase JWT authentication to a language-model endpoint that validates every response against a **Zod** schema, retries within set limits, and quarantines malformed output so callers always get clean JSON or a clear error.

PROJECTS

Diacify - Clinical Decision Support System

Aug 2026 - Sep 2026

React 19, Node.js, Express.js, Supabase, Zod, scikit-learn | diacify.vercel.app

- Rebuilt a clinical decision support tool in **React** and **Node.js**, merging a three-service prototype into one application so a clinician records a patient visit and gets a diabetes risk score back in the same request.
- Secured every database query behind **Supabase** Auth and PostgreSQL row-level security so no clinician can read another clinician's patients, and shared one **Zod** validation schema between browser and server so both sides always agree on what valid patient data looks like.
- Trained a Random Forest classifier in Python with **scikit-learn** on records from 662 patients across 13 clinical indicators, reaching 94% test accuracy aligned with published diabetes diagnostic thresholds so the score a clinician sees reflects recognised clinical guidance rather than an arbitrary cutoff.

ClaimLens - AI-Powered Damage Claim Triage

Jul 2026

Python, FastAPI, Anthropic API, Next.js 15, TypeScript | github.com/deshanekanayaka/claimlens

- Built an insurance claim triage tool in Python and FastAPI with a **Next.js** and TypeScript interface that reads a damage claim alongside its photographs and recommends approve, reject, or escalate, streaming each step live so a reviewer sees how a recommendation was reached rather than a bare verdict.
- Split the decision across three separate **Anthropic Claude** calls for claim extraction, image analysis, and final verdict, with each step able to stop and return 'not enough information' rather than force a guess, because a wrong automated rejection on a genuine claim costs far more than asking a person to look.

EDUCATION

University of Westminster

London, UK

BSc (Hons) Computer Science, First Class Honours

Sep 2023 - Jun 2026

- **Coursework:** Object-Oriented Programming, Algorithms, Client-Server Architectures, Cybersecurity, Artificial Intelligence

TECHNICAL SKILLS

Languages: Python, TypeScript

Frameworks: React, Next.js, Node.js, Express.js, FastAPI

Data and ML: PostgreSQL, Supabase, scikit-learn, Anthropic and OpenAI APIs

Infrastructure and Tooling: Zod, Pydantic